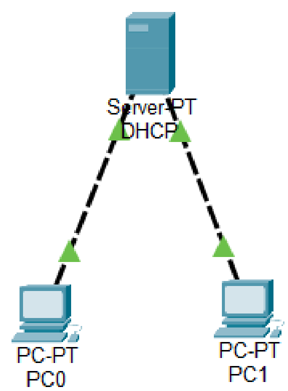
DAY-2

❖ **DHCP Configuration Theory (30 min):DHCP process: Discover, Offer, Request, Acknowledge. Practical (30 min):Configure a DHCP server in Cisco Packet Tracer. Task: Create a diagram showing the DHCP process flow.**

1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

ANS :



DHCP Server Configuration

- **Desktop > IP Configuration**
    - IP Address: **192.168.1.2**
    - Subnet Mask: **255.255.255.0**
    - Default Gateway: **192.168.1.1**

- **Services > DHCP**
    - DHCP Service: **ON**
    - Pool Name: **Office**
    - Default Gateway: **192.168.1.1**
    - DNS Server: **8.8.8.8**
    - Start IP Address: **192.168.1.100**
    - Subnet Mask: **255.255.255.0**

- Maximum Users: **50**
- Click **Add**

➢ **PC Configuration**

On **PC0**

- Desktop → IP Configuration
- Select **DHCP**

Assigned IP:

- **192.168.1.100**
- Subnet Mask: **255.255.255.0**
- Gateway: **192.168.1.1**

On **PC1**

- Desktop → IP Configuration
- Select **DHCP**

Assigned IP:

- **192.168.1.101**
- Subnet Mask: **255.255.255.0**
- Gateway: **192.168.1.1**

➢ **Verification**

Both PCs received different IP addresses automatically from the DHCP server.

2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint : Use arrows to show the direction of each message between client and server.

ANS :

```
Client PC                                   DHCP Server
    |                                             |
    |------ DHCP Discover --------------------->|
    |                                             |
    |<------- DHCP Offer ------------------------|
    |                                             |
    |------ DHCP Request ---------------------->|
    |                                             |
    |<----- DHCP Acknowledge (ACK) --------|
    |                                             |
    |     IP Address Assigned                    |
```

- **DHCP Process Explanation**

1. Discover

- Client broadcasts a request to find available DHCP servers.

2. Offer

- DHCP server offers an available IP address.

3. Request

- Client requests the offered IP address.

4. Acknowledge (ACK)

- DHCP server confirms and leases the IP address to the client.

3. **Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.**

ANS :

- **Configuration**

  DHCP Pool

  - Start IP Address: **192.168.1.100**
  - Maximum Users: **1**

  Now connect:

  - PC0
  - PC1

  Both are set to **DHCP**.

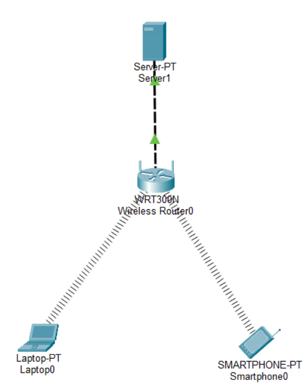
- **Result**

Device	Assigned IP
PC0	192.168.1.100
PC1	No IP (169.254.x.x or Request Failed)

4. **Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.**

ANS :

Network Diagram



- **DHCP Configuration**

Server IP

- **192.168.10.2**

DHCP Pool

- Pool Name: Restaurant
- Gateway: **192.168.10.1**
- DNS: **8.8.8.8**
- Start IP: **192.168.10.100**
- Maximum Users: **20**

Assigned IP Addresses

Device	**Assigned IP**
Laptop	192.168.10.100
Smartphone (Generic PC)	192.168.10.101

Why DHCP is Useful

DHCP automatically assigns IP addresses to all customer devices when they connect to the restaurant's Wi-Fi. This eliminates manual configuration, reduces network errors, and allows many users to connect quickly and efficiently.